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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_0^{\frac{\pi}{2}} \sin x dx$, by dividing the interval into four equal parts, width of the interval should be

Answer (Please select your correct option)

- π
- $\frac{\pi}{2}$
- $\frac{\pi}{4}$
- $\frac{\pi}{8}$ R

We shall at first divide the range of integration $(0, \pi)$ into six equal parts so that each part is of width $\pi/6$ and write down the table of values:

X	0	π/6	π/3	π/2	2π/3	5π/6	π
Y=sinx	0.0	0.5	0.8660	1.0	0.8660	0.5	0.0

✓

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Lagrange's interpolation formula is used when the values of the independent variable are

Answer (Please select your correct option)

- Equally spaced
- Not equally spaced
- Constant
- None of the above

R ✓

LAGRANGE'S INTERPOLATION FORMULA

Newton's interpolation formulae developed earlier can be used only when the values of the independent variable x are equally spaced. Also the differences of y must ultimately become small.

If the values of the independent variable are not given at equidistant intervals, then we have the basic formula associated with the name of Lagrange which will be derived now.

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are (n+1) values of y corresponding to (n+1) values of x, then we can represent the function f(x) by a polynomial of degree

Answer (Please select your correct option)

- n+2
- n+1
- n

R ✓

, x_1, ...x_n . Since there are $(n + 1)$ values of y corresponding to $(n + 1)$ values of x, we can represent the function $f(x)$ by a polynomial of degree n.

- n-1

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	1	3	7
$f(x)$	2	4	10

Answer (Please select your correct option)

- Newton's forward difference interpolation formula
- Newton's backward difference interpolation formula
- Lagrange's interpolation formula
- None of the given choices

R

LAGRANGE'S INTERPOLATION FORMULA

Newton's interpolation formulae developed earlier can be used only when the values of the independent variable x are equally spaced. Also the differences of y must ultimately become small.

If the values of the independent variable are not given at equidistant intervals, then we have the basic formula associated with the name of Lagrange which will be derived now.

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	0	1	4
$y = f(x)$	2	1	4

Using Newton's divided difference formula, find the quadratic equation for the following

X	0	1	4
Y	2	1	4

data.
Hence find y (2).

Answer (Please select your correct option)

- -1
- 1
- 2
- 4

Solution:
The divided difference table for the given data is constructed as:

X	Y	1st D.D	2nd D.D
0	2		
1	1	-1	1/2
4	4	1	

Now, using Newton's divided difference formula, we have

$$y = 2 + (x-0)(-1) + (x-0)(x-1)\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}(x^2 - 3x + 4)$$

Hence, $y(2) = 1.$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Relationship between differential operator and backward difference operator is given by

Answer (Please select your correct option)

- $hD = -\log(1+\nabla)$
- $hD = \log(1+\nabla)$
- $hD = \log(1-\nabla)$
- $hD = -\log(1-\nabla)$ R ✓

Using backward difference operator , we have $hD = -\log(1-\nabla)$.

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Shift operator is defined as

Answer (Please select your correct option)

- $Ef(x) = f(x-h)$
- $Ef(x) = -f(x-h)$
- $Ef(x) = -f(x+h)$
- $Ef(x) = f(x+h)$ ✓ R

Shift operator, E

Let $y = f(x)$ be a function of x, and let x takes the consecutive values x, $x + h$, $x + 2h$, etc.
We then define an operator having the property

$$E f(x) = f(x+h)$$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

using two-point equation the value of f '(0.5) is.....:

Answer (Please select your correct option)

- 0.73
- -0.75
- 0.5
- -0.5

R ✓

$$f'(0.1) = \frac{f(0.2) - f(0.1)}{h} = \frac{0.475 - 0.425}{0.1} = \frac{0.050}{0.1} = 0.5.$$

$$f'(0.2) = \frac{f(0.3) - f(0.2)}{h} = \frac{0.400 - 0.475}{0.1} = \frac{-0.075}{0.1} = -0.75.$$

$$f'(0.3) = \frac{f(0.4) - f(0.3)}{h} = \frac{0.450 - 0.400}{0.1} = \frac{0.05}{0.1} = 0.5.$$

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

MCQ INCOMPLETE

$f''(0.5)$ **using three-point equation will be calculated as .**

Answer (Please select your correct option)

- 17.5
- 12.5
- 7.5 **R** ✓
- -2.3

$$f''(0.3) = \frac{f(0.4) - 2f(0.3) + f(0.2)}{h^2 = 0.01} = \frac{0.125}{0.01} = 12.5$$

$$f''(0.4) = \frac{f(0.5) - 2f(0.4) + f(0.3)}{h^2 = 0.01} = \frac{0.025}{0.01} = 2.5$$

$$f''(0.5) = \frac{f(0.6) - 2f(0.5) + f(0.4)}{0.01} = \frac{0.075}{0.01} = 7.5$$

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

The magnitude of truncation error in Milne's predictor formula is

Answer (Please select your correct option)

- $\frac{1}{90}h\Delta^4 y_0'$
- $\frac{1}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^4 y_0'$ R ✓

From these equations, we observe that the magnitude of the truncation error in corrector formula is $1/90h\Delta^4 y_0'$ while the truncation error in predictor formula is $28/90h\Delta^4 y_0'$

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Let I denotes the closed interval spanned by $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, \overline{x}$. Then $F(x)$ vanishes ------times in the interval I.

Answer (Please select your correct option)

- 6
- 9 **R** ✓ Let I denotes the closed interval spanned by the values $x_0, \ldots, x_n, \overline{x}$. Then $F(x)$ vanishes $(n+2)$ times in the interval I.
- 7
- 8

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

From the following table of values:

x	1.3	1.5	1.7	1.9	2.1	2.3
y	2.9648	2.6599	2.3333	1.9922	1.6442	1.2969

Estimation of $y'(1.3)$ will be done using the formula of differential operator in terms of

Answer (Please select your correct option)

- Backward difference operator
- Central difference operator
- None of the given choices
- Forward difference operator

R

Forward Differences

For a given table of values $(x_k, y_k), k = 0,1,2,...,n$ with equally spaced abscissas of a function $y = f(x)$, we define the forward difference operator Δ as follows

$\Delta y_i = y_{i+1} - y_i, \quad i = 0,1,...,(n-1)$

To be explicit, we write

$$\Delta y_0 = y_1 - y_0$$
$$\Delta y_1 = y_2 - y_1$$
$$\vdots \quad \vdots \quad \vdots$$
$$\Delta y_{n-1} = y_n - y_{n-1}$$

✓

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Trapezoidal rule, we approximate f(x) in the form .....

Answer (Please select your correct option)

- $ax+b$
- ax^2+bx+c
- ax^3+bx^2+cx+d
- $ax^4+bx^3+cx^2+dx+e$

R ✓

The Trapezoidal Rule (Composite Form)

The Newton-Cotes formula is based on approximating y = f (x) between (x_0, y_0) and (x_1, y_1) by a straight line, thus forming a trapezium, is called trapezoidal rule. In order to evaluate the definite integral

$$I = \int_a^b f(x)dx$$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Simpson's 3/8 rule, we approximate f(x) in the form .....

Answer (Please select your correct option)

- $ax+b$
- ax^2+bx+c
- ax^3+bx^2+cx+d R ✓
- $ax^4+bx^3+cx^2+dx+e$

Thus Simpson's 1/3 rule is based on fitting three points with a quadratic.
Similarly, for n = 3, the integration is found to be

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0+3y_1+3y_2+3y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Two segment trapezoidal rule of integration is exact for integrating at most ..... order polynomial.

Answer (Please select your correct option)

- first

R ✓

- second
- third
- fourth

The Trapezoidal Rule (Composite Form)

The Newton-Cotes formula is based on approximating $y = f(x)$ between (x_0, y_0) and (x_1, y_1) by a straight line, thus forming a trapezium, is called trapezoidal rule. In order to evaluate the definite integral

$$I = \int_a^b f(x)dx$$

The single segment trapezoidal rule is exact for at most a first order polynomial. The two segment trapezoidal rule is also exact only for the same order of polynomial, that is, a first order polynomial.

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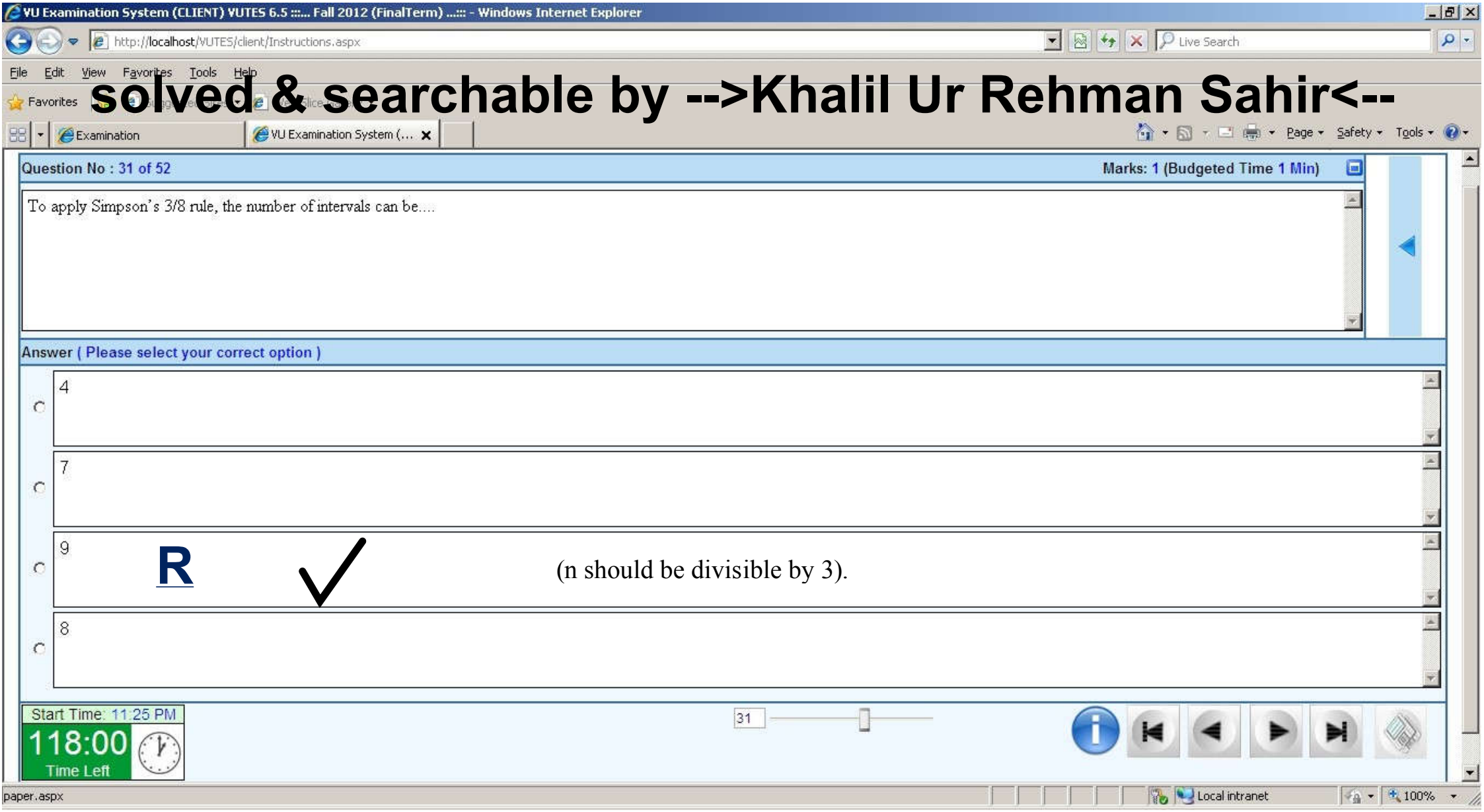
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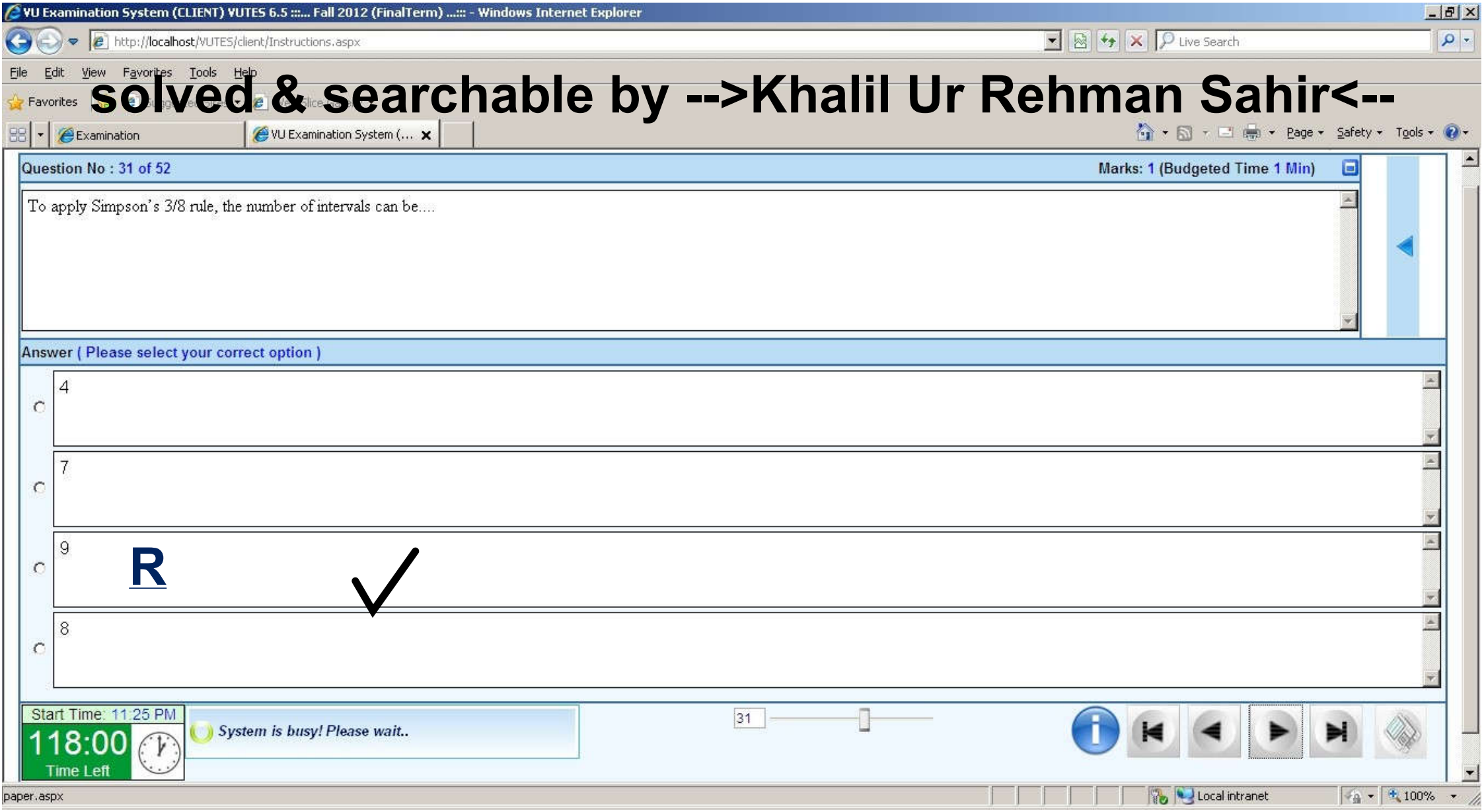
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Question No : 31 of 52
Marks: 1 (Budgeted Time 1 Min)
To apply Simpson's 3/8 rule, the number of intervals can be....
Answer (Please select your correct option)
4
7
9
R
8
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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Romberg's integration method is ---------- than Trapezoidal and Simpson's rule.

Answer (Please select your correct option)

- none of the given choices
- more accurate
- less accurate
- equally accurate

R ✓

ROMBERG'S INTEGRATION

We have observed that the trapezoidal rule of integration of a definite integral is of $O(h^2)$, while that of Simpson's 1/3 and 3/8 rules are of fourth-order accurate.

We can improve the accuracy of trapezoidal and Simpson's rules using Richardson's extrapolation procedure which is also called Romberg's integration method.

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The global error in Simpson's 3/8 rule is of

Answer (Please select your correct option)

- $o(h^2)$
- $o(h^3)$
- $o(h^4)$ R ✓
- None of these choices

in Simpson's 1/3 rule, the global error is of $O(h^4)$.

We obtain the composite form of Simpson's 3/8 rule as

$$\int_a^b f(x)dx = \frac{3}{8}h[y(a)+3y_1+3y_2+2y_3+3y_4+3y_5+2y_6+\cdots$$
$$+2y_{n-3}+3y_{n-2}+3y_{n-1}+y(b)]$$

With the global error E given by

$$E = -\frac{x_n - x_0}{80}h^4 y^{(iv)}(\xi)$$

It may be noted that the global error in Simpson's 1/3 and 3/8 rules are of the same order.

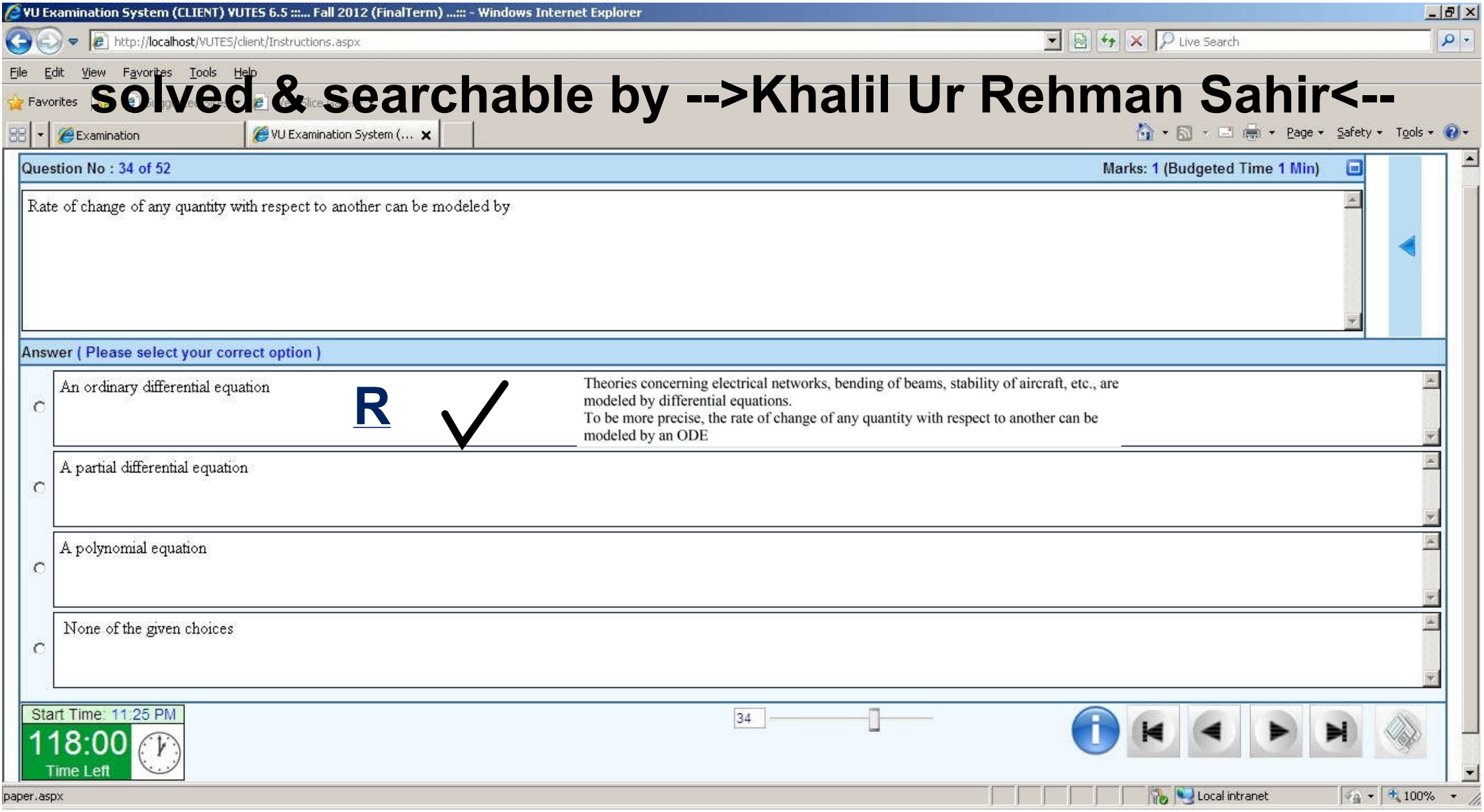
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Question No : 34 of 52
Marks: 1 (Budgeted Time 1 Min)
Rate of change of any quantity with respect to another can be modeled by
Answer (Please select your correct option)
An ordinary differential equation
R
Theories concerning electrical networks, bending of beams, stability of aircraft, etc., are modeled by differential equations.
To be more precise, the rate of change of any quantity with respect to another can be modeled by an ODE
A partial differential equation
A polynomial equation
None of the given choices
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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

A third order ordinary differential equation can be reduced to a system of ---------- first order ordinary differential equations.

Answer (Please select your correct option)

0

1

2

3

R

For example, consider a second order differential equation of the form

$$y'' = f(t, y, y')$$

Introducing the substitution $p = y'$, the above equation reduces to a system of two first order differential equations, such as

$$y' = p, \qquad p' = f(t, y, p)$$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.02 at t=0.02. Using Euler's method, y at t= 0.04, h=0.02 is

h = t1 - t0

Answer (Please select your correct option)

- 3.0392

$$y_1 = y_0 + hf(t_0, y_0),$$

- 2.0392

$$y_2 = y_1 + hf(t_1, y_1) = 1.02 + 0.02\frac{1.02-0.02}{1.02+0.02} = 1.0392$$

- 1.0392 **R** ✓
- 0.0392

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_1 is given by

Answer (Please select your correct option)

- $k_1 = hf(x_n, y_n)$ R ✓
- $k_1 = 2hf(x_n, y_n)$
- $k_1 = 3hf(x_n, y_n)$
- None of the given choices

The fourth-order R-K method was described as

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_4 is given by

Answer (Please select your correct option)

- $k_4 = hf(x_n + 2h, y_n + 2k_3)$
- $k_4 = hf(x_n - h, y_n - k_3)$
- $k_4 = hf(x_n + h, y_n + k_3)$ R ✓
- None of the given choices

The fourth-order R-K method was described as

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The truncation error in Adam's predictor formula is

Answer (Please select your correct option)

- $\frac{251}{720}h\nabla^4 y_n'$ **R** This is known as Adam's predictor formula. The truncation error is $(251/720)h\nabla^4 y_n'$. ✓
- $\frac{251}{720}h\nabla^4 y_{n+1}'$
- $\frac{19}{720}h\nabla^4 y_{n+1}'$
- $\frac{19}{720}h\nabla^4 y_n'$

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x^2 + 2y \; ; \; y(1) = 3$$

$h = 1$, By Euler's method $y(2)$ is calculated as

Solution:

Given $y' = 2y + x^2, y(1) = 3, h = 1, y(2) = ?$

Answer (Please select your correct option)

Euler method

- 4
- 6
- 8
- 10 R ✓

$$y_1 = y_0 + hf\left(x_0, y_0\right) = 3 + (1)f(1, 3) = 3 + (1) \cdot (7) = 3 + (7) = 10$$

$$\therefore y(2) = 10$$

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Obtain numerically the solution of

$$y' = x + y^2,\ y(0) = 2$$

Using Euler's method to find y at $x = 1$, h=1

Answer (Please click here to Add Answer)

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Solution:

Given $y' = x + y^2, y(0) = 2, h = 1, y(1) = ?$

Euler method

$$y_1 = y_0 + hf\left(x_0, y_0\right) = 2 + (1)f(0, 2) = 2 + (1) \cdot (4) = 2 + (4) = 6$$

$$\therefore y(1) = 6$$

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a backward difference formula of $D^2 f(x)$.

Answer (Please click here to Add Answer)

$$\frac{d}{dx} y_n = Dy_n = y_n'$$

$$= \frac{1}{h}\left(\nabla y_n - \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4} + \cdots\right)$$

$$y_n'' = D^2 y_n$$

$$= \frac{1}{h^2}\left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \cdots\right)$$

$$D = \frac{1}{h}\left(\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots\right)$$

$$D^2 = \frac{1}{h^2}\left(\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots\right)^2$$

$$= \frac{1}{h^2}\left(\nabla^2 + \nabla^3 + \frac{11}{12}\nabla^4 + \frac{5}{6}\nabla^5 + \cdots\right)$$

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a formula for Simpson's 1/3 rule.

$$\int_{x_0}^{x_2} f(x)dx = x_0y_0 + x_1y_1 + x_2y_2 + \text{Error} = \frac{h}{3}(y_0 + 4y_1 + y_2) - \frac{h^5}{90}y^{(iv)}(\xi)$$

Answer (Please click here to Add Answer)

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$$\int_{x_0}^{x_{2N}} f(x)dx = \frac{h}{3}[y_0 + 4(y_1 + y_3 + \cdots + y_{2N-1}) + 2(y_2 + y_4 + \cdots + y_{2N-2}) + y_{2N}] + \text{Error term}$$

This formula is called composite Simpson's 1/3 rule. The error term E, which is also called global error, is given by

$$E = -\frac{N}{90}h^5y^{(iv)}(\xi) = -\frac{x_{2N} - x_0}{180}h^4y^{(iv)}(\xi)$$

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Evaluate the integral

$$\int_3^5 (\log x + 2)\,dx$$

Using Simpson's 3/8 rule

Take h=1

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

Answer (Please click here to Add Answer)

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h=1

x= 3 4 5

logx + 2= 2.4771 2.6020 2.6989

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Runge-Kutta Method of order four to find the values of
k_1 and k_2 for the initial value problem

$$y' = yx^2, y(0) = 1, h = 0.1$$

Answer (Please click here to Add Answer)

Solution:

Given $y' = yx^2, y(0) = 1, h = 0.1$

Forth order R-K method

$$k_1 = hf\left(x_0, y_0\right) = (0.1)f(0, 1) = (0.1) \cdot (0) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.1)f(0.05, 1) = (0.1) \cdot (0.0025) = 0.0003$$

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

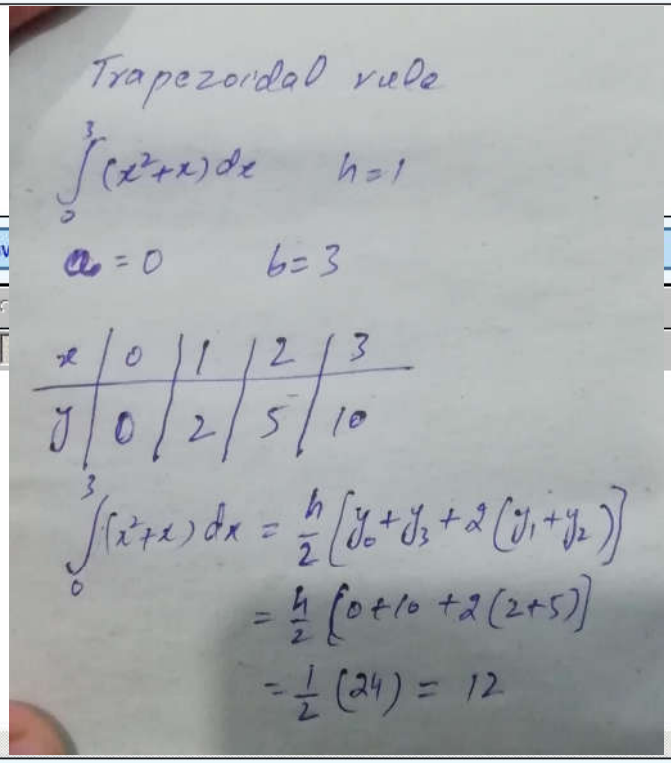
Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Trapezoidal rule
Take h=1

Answer (Please click here to Add Answ

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x+1)dx$$

Using Simpson's 3/8 rule

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

Answer (Please click here to Add Answer)

Using Simpson's $\frac{3}{8}$ Rule

x	0	1	2	3
y	1	2	3	4

$$\int y dx = \frac{3h}{8}\left[(y_0 + y_3) + 2() + 3(y_1 + y_2)\right]$$

$$\int y dx = \frac{3 \times 1}{8}[(1 + 4) + 2 \times () + 3 \times (2 + 3)]$$

$$\int y dx = \frac{3 \times 1}{8}[(1 + 4) + 2 \times (0) + 3 \times (5)]$$

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

From the following table of values, construct backward difference table.

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Answer (Please click here to Add Answer)

Newton's backward differentiation table is

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
0.1	1.10517			
		0.11623		
0.2	1.2214		0.01223	
		0.12846		**0.00127**
0.3	1.34986		**0.0135**	
		0.14196		
0.4	**1.49182**			

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^4 (x^2 + 1)dx$$

Using Simpson's 1/3 rule
Take h=1

Answer (Please click here to Add Answer)

x1 = 0 and x2 = 4
Step value (h) = 1

Solution:
Equation is $f(x) = x^2 + 1$.

The value of table for x and y

x	0	1	2	3	4
y	1	2	5	10	17

$$\int ydx = \frac{1}{3}[(1 + 17) + 4 \times (2 + 10) + 2 \times (5)]$$

$$\int ydx = \frac{1}{3}[(1 + 17) + 4 \times (12) + 2 \times (5)]$$

$$\int ydx = 25.3333$$

Solution by Simpson's $\frac{1}{3}$ Rule is 25.3333

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